

Question Paper

Exam Date & Time: 22-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Cloud Computing [IT_4441]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Identify and explain the fundamental differences between IaaS, PaaS, and SaaS and why the insurance company INSUREME needs to choose them for their cloud migration. (5)

A)

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Specify and explain the reasons why Platform as a Service (PaaS) the preferred environment for a cost-conscious e-commerce startup is that prioritizes speed and developer focus when launching a new web application. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) List the various components of the service-oriented architectures and its uses. (2)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 2) In a cloud computing environment, three users (User P, User Q, and User R) are submitting tasks to a shared resource pool. The system uses a fair queueing algorithm to allocate resources. Each user submits tasks at different rates and requires different amounts of resources. (5)

A)

- User P submits tasks at a rate of 4 tasks per minute, each requiring 2 units of resource.
- User Q submits tasks at a rate of 5 tasks per minute, each requiring 3 units of resource.
- User R submits tasks at a rate of 3 tasks per minute, each requiring 1 unit of resource.

The total available resource in the system is 22 units per minute. Based on the above scenario the following questions.

1. Calculate the total resource demand per minute for each user.
2. Determine the fair share of resources each user should receive per minute based on their demand and the total available resources.
3. Calculate the number of tasks each user can complete per minute with their fair share of resources.
4. Evaluate the system's efficiency by comparing the total resource utilization with the available resources.

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) Analyze the fundamental differences in network addressability concept between a Virtual NIC and a Physical NIC within a Virtualized Data Center network infrastructure. Explain in detail. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) Identify the two distinct resource allocation requirements that every scheduling policy must address (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) Define Network Virtualization in the context of a Virtualized Data Centre(VDC). List the various components used and explain key actions that network virtualization enables for independent virtual networks. (5)

A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Identify and explain the benefits that an organization gains from implementing VLAN technology in their network infrastructure. Also explain how the VLAN are configured. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) list the key componets & their functions available in the OpenStack tool. (2)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 4) Identify the two primary goals that Redundant Array of Independent Disks (RAID) is designed to achieve regarding data access and system availability. Explain the concepts of Striping and Redundancy, detailing which primary goal each mechanism addresses. (5)

A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Compare the pros and cons of using both synchronous and asynchronous replication when setting up remote data recovery in a VDC environment (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) Mention how SOA Communications are performed using SOAP and REST. (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 5) Imagine Manipal hospital is setting up its patient records system. To make sure the system is secure, they must meet the three basic goals of the CIA Triad (Confidentiality, Integrity, and Availability). Briefly define each of the three goals by explaining what specific security need it addresses in the context of handling sensitive patient records (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Explain the keyways Multi-Factor Authentication (MFA) improves security by requiring more than one proof of identity in detail. (5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4441:Cloud Computing

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	6
3	3	CO 3	5
4	4	CO 4	19
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4441:Cloud Computing

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	7	25(50%)
4	Analysis	7	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	14	50 (100%)